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Multivariate Gaussian RVs

$U_1, U_2, \dots, U_n \sim$ iid $N(0,1)$

$\underline{U} = \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix}$ vector of RVs

$$f_{U_1 U_2 \dots U_n}(u_1, u_2, \dots, u_n) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-u_i^2/2}$$

$\mu_{U_i} = 0, \quad \sigma_{U_i} = 1, \quad \rho_{U_i U_j} = 0 \quad i \neq j$

X, Y: jointly Gaussian

$\mu_X, \sigma_X^2, \mu_Y, \sigma_Y^2, \rho_{XY}$

$U, V \sim$ iid $N(0,1)$

$$\frac{X - \mu_X}{\sigma_X} = U$$

$$\frac{Y - \mu_Y}{\sigma_Y} = \rho U + \sqrt{1-\rho^2}\, V$$

$$\underline{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} = A \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix} = A\underline{U}$$

A: $n \times n$ matrix

$\underline{X}$: multivariate jointly Gaussian

→ any linear combination is Gaussian.

$$\underline{X} = A\underline{U} + \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_n \end{bmatrix} \quad \underline{\mu}$$

$$\underline{X} = A\underline{U} + \underline{\mu}$$

$$X_i = a_{i1} U_1 + \dots + a_{in} U_n + \mu_i$$

$$\mu_{X_i} = \mu_i \qquad \sigma_{X_i}^2 = a_{i1}^2 + \dots + a_{in}^2$$

$$E\left[(X_i - \mu_i)(X_j - \mu_j)\right] = \begin{cases} a_{i1}^2 + \dots + a_{in}^2 & i = j \\ a_{i1}a_{j1} + \dots + a_{in}a_{jn} & i \neq j \end{cases}$$

$\underline{X} - \underline{\mu}$ → $\begin{bmatrix} X_1 - \mu_1 \\ X_2 - \mu_2 \\ \vdots \\ X_n - \mu_n \end{bmatrix}$ $\quad (\underline{X} - \underline{\mu})^T$ → $\begin{bmatrix} X_1 - \mu_1 & X_2 - \mu_2 & \dots & X_n - \mu_n \end{bmatrix}$

$$= (\underline{X} - \underline{\mu})(\underline{X} - \underline{\mu})^T$$

$$\Sigma_{\underline{X}} = E\left[(\underline{X}-\underline{\mu})(\underline{X}-\underline{\mu})^T\right] = AA^T$$ "Covariance matrix"

$$\mathbb{R} \ni \underline{x}^T A A^T \underline{x} = (A^T \underline{x})^T A^T \underline{x} \geq 0$$

$\underline{\mu} = 0$

$$\underline{X} = A\,\underline{U} \qquad \Sigma_{\underline{X}} = AA^T \qquad \det(\Sigma_{\underline{X}}) = \det(A)^2$$

A: invertible

$$f_{\underline{X}}(\underline{x}) = f_{\underline{U}}(A^{-1}\underline{x}) \frac{1}{|\det(A)|}$$

$$f_{\underline{U}}(\underline{u}) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-u_i^2/2}$$

$$= \frac{1}{(2\pi)^{n/2}} e^{-\frac{\underline{u}^T\underline{u}}{2}}$$

$$= \frac{1}{(2\pi)^{n/2}|\det(A)|} \cdot e^{-\frac{\underline{x}^T (A^{-1})^T A^{-1} \underline{x}}{2}}$$

$$= \frac{1}{(2\pi)^{n/2}\sqrt{\det(\Sigma_{\underline{X}})}} \cdot e^{-\frac{\underline{x}^T \Sigma_{\underline{X}}^{-1} \underline{x}}{2}}$$

Multivariate Gaussian Density

Given

$\underline{X}$: $\mu_{\underline{X}}$ & $\Sigma_{\underline{X}}$.

Is there an 'A' s.t. $A\underline{X} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix} \sim$ iid $N(0,1)$?

YES

$\underline{v}_i$: e.vec with e.val. λ_i

$$\Sigma_{\underline{X}}\,\underline{v}_i = \lambda_i \underline{v}_i$$

$i = 1, \ldots, n$

& $\underline{v}_i^T \underline{v}_j = \begin{cases} 0, & \text{if } i \neq j \\ 1, & \text{if } i = j \end{cases}$

$$\Sigma_{\underline{X}} \begin{bmatrix} | & | & & | \\ \underline{v}_1 & \underline{v}_2 & \cdots & \underline{v}_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \underline{v}_1 & \underline{v}_2 & \cdots & \underline{v}_n \\ | & | & & | \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix}$$

V $\qquad$ V

$$VV^T = I_n$$

$$\Sigma_{\underline{x}} = V \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} V^T$$

$$f_{\underline{x}}(\underline{x}) = \frac{1}{(2\pi)^{n/2} |\det(\Sigma_{\underline{x}})|^{1/2}} e^{-\frac{\underline{x}^T (VDV^T)^{-1} \underline{x}}{2}}$$

$$\underline{x}^T (V\sqrt{D})^T (V\sqrt{D})^{-1} \underline{x}$$

$$\underline{u}^T \qquad \underline{u}$$